Freight Management System

End-to-End Logistics & Freight Quote Management Platform

Project Type	Logistics & Freight Management System
Industry	Industrial Machinery Transportation
Role	Full Stack MERN Developer

Frontend	React.js, JavaScript
Backend	Node.js, Express.js, REST APIs
Database	MongoDB, Mongoose
Communication	Nodemailer, ClickSend SMS
Integrations	Google Maps API
Infrastructure	DigitalOcean, Nginx, PM2, GitHub Actions

Project Overview

The Freight Management System is a complete logistics management solution developed to manage the transportation of industrial machinery from the initial freight request through company selection, quotation, acceptance, delivery, and customer communication.

The system was designed around the specific requirements of machinery transportation, where every shipment can contain detailed machine information, pickup and delivery locations, transportation requirements, required dates, delivery instructions, and freight company assignments.

Instead of managing freight requests manually through emails and spreadsheets, the platform provides a centralized workflow for customers, freight companies, and administrators.

The complete workflow is:

Freight Job → Freight Companies → Quotes → Quote Selection → Assignment → Notifications → Pickup → Delivery → Tracking

The Problem

Transporting industrial machinery involves more complexity than standard parcel shipping.

A single freight request may require:

- Specific machine information
- Machine dimensions and weight
- Pickup location
- Delivery location
- Required delivery date
- Delivery notes
- Freight company selection
- Multiple transportation quotes
- Quote comparison
- Quote acceptance
- Company communication
- Pickup and delivery tracking

Managing these processes manually can result in delays, missed communication, duplicate work, and difficulty tracking the current status of a shipment.

The goal was to build a centralized digital workflow that could automate these processes.

Solution

The platform provides a complete freight lifecycle management system.

```
Customer / Admin
       |
Create Freight Job
       |
Add Machine Details
       |
Pickup & Delivery Information
       |
Notify Freight Companies
       |
Freight Companies Submit Quotes
       |
Review Quotes
       |
Accept Best Quote
       |
Assign Freight Company
       |
Automated Notifications
       |
Pickup / Transit
       |
Delivery
       |
Shipment Tracking
```

This creates a structured workflow from freight creation to final delivery.

1. Freight Job Creation

The system starts with the creation of a freight job. A freight job contains the information required to transport a particular machine. Important information can include:

- Machine details
- Machine specifications
- Pickup location
- Delivery location
- Required date
- Delivery requirements
- Delivery notes
- Customer information
- Freight requirements

This creates a single source of truth for the entire shipment.

2. Machine-Specific Freight Information

Because the platform is designed for industrial machinery transportation, the freight system handles machine-specific information rather than treating every shipment as a generic package. A freight job can be associated with details such as:

```
Machine
+-- Name
+-- Model
+-- Dimensions
+-- Weight
+-- Quantity
+-- Transportation Requirements
```

This information is important for freight companies when calculating transportation requirements and submitting accurate quotes.

3. Pickup & Delivery Management

Each freight job contains structured pickup and delivery information.

```
PICKUP
Location / Contact
Required Date / Instructions
       |
TRANSPORTATION
       |
DELIVERY
Location / Contact
Required Date / Instructions
```

This allows freight companies to understand exactly where the machinery needs to be collected and delivered.

4. Freight Company Registration

Freight companies can register with the platform to participate in transportation jobs. The registration workflow allows the business to maintain freight company information and control which companies can participate in the quoting process. The system can manage company information such as:

- Company name
- Contact information
- Registration information
- Approval status
- Operational details

This creates a controlled network of transportation providers.

5. Freight Company Approval

New freight companies do not automatically become active participants. The platform includes an approval workflow where companies can be reviewed before receiving freight opportunities.

```
Company Registration
        |
Review
        |
Approved / Rejected
        |
Approved Company
        |
Can Participate in Freight Jobs
```

This helps maintain control over the freight-provider network.

6. Freight Job Browsing

Approved freight companies can browse available freight jobs. They can review relevant job information before deciding whether they want to submit a quotation. The job information provides the context required to determine whether the shipment is suitable for the company's transportation capabilities.

7. Quotation Management

One of the central features of the system is the freight quotation workflow. Multiple freight companies can submit quotes for the same freight job. For example:

```
Freight Job #FR-1024

Company A -> $1,250
Company B -> $1,180
Company C -> $1,350
Company D -> $1,220
```

The customer or administrator can then review the available quotes and select the most appropriate option.

8. Quote Submission

Freight companies can submit their own quotation for a job. A quote is associated with:

- Freight job
- Freight company
- Quoted amount
- Relevant details

- Quote status
- Submission information

This creates a structured relationship between the shipment and transportation provider.

9. Quote Acceptance

Once quotes have been received, the customer/admin can accept the selected quotation. The workflow becomes:

```
Multiple Quotes
      |
Review
      |
Select Quote
      |
Accept Quote
      |
Assign Freight Company
      |
Update Freight Status
```

The accepted quote becomes the active freight arrangement for the job.

10. Automatic Quote Rejection

A key automation feature is the handling of unsuccessful quotes. When one quote is accepted, other pending quotes can automatically be rejected. For example:

```
Company A -> Accepted [OK]

Company B -> Automatically Rejected
Company C -> Automatically Rejected
Company D -> Automatically Rejected
```

This eliminates the need for administrators to manually update every unsuccessful quotation. It also ensures that the freight job has a clear final quote status.

11. Quote Status Management

The system manages quote states throughout the lifecycle. Typical states include:

```
Pending
   |
Accepted
   OR
Rejected
```

This makes it easier for both administrators and freight companies to understand the current state of each quotation.

12. Automated Email Notifications

Email notifications are integrated into important freight events. Notifications can be triggered when:

- A freight job becomes available
- A freight company submits a quote

- A quote is accepted
- A quote is rejected
- A freight job is assigned
- A reminder is required
- A shipment status changes

This reduces manual communication between the parties involved.

13. Automated SMS Notifications

SMS communication was integrated using **ClickSend**. This provides an additional communication channel for important freight updates. For example:

```
Freight Job
    |
Status Change
    |
System Event
    |
ClickSend
    |
SMS Notification
```

This is particularly useful for time-sensitive transportation updates.

14. Email + SMS Communication Workflow

The system combines email and SMS communication to keep stakeholders informed.

```
                Freight Event
                     |
             Notification Service
                /            \
               /              \
           Email               SMS
          Nodemailer        ClickSend
```

This reduces the dependency on manual follow-up.

15. Automated Reminders

The system can also support automated reminders around freight activities. This is useful for ensuring important actions do not remain pending indefinitely. Examples include:

- Quote reminders
- Freight status reminders
- Delivery-related communication
- Pending action notifications

Automation helps reduce operational overhead.

16. Freight Status Lifecycle

Freight jobs follow a structured status lifecycle. Conceptually:

```
Created
   |
Available for Quotes
   |
Quotes Received
   |
Quote Accepted
   |
Freight Company Assigned
   |
Pickup
   |
In Transit
   |
Delivered
```

This gives administrators and stakeholders a clear view of where each shipment currently stands.

17. Real-Time Shipment Tracking

The system supports shipment tracking functionality so freight progress can be monitored through the platform. Tracking provides visibility into the movement/status of freight rather than requiring customers to rely entirely on manual communication.

```
Freight Job
    |
Assigned
    |
Pickup
    |
In Transit
    |
Delivery
    |
Completed
```

This provides greater transparency across the logistics workflow.

18. Google Maps Integration

Google Maps API is used as part of the location and logistics workflow. The integration supports location-based functionality around:

- Pickup locations
- Delivery locations
- Route information
- Freight location context

This provides a more useful geographic view of freight operations.

19. Freight Job Dashboard

The system provides a centralized view of freight operations. Administrators can monitor:

- Active freight jobs
- Pending jobs
- Quotes

- Accepted quotes
- Freight companies
- Assigned shipments
- Delivery status
- Notifications

This reduces the need to manage logistics operations across disconnected tools.

20. Freight Company Dashboard

Freight companies have a workflow focused on the jobs available to them. They can:

```
View Available Jobs
        |
Review Machine Details
        |
Review Pickup / Delivery
        |
Submit Quote
        |
Track Quote Status
        |
Manage Accepted Jobs
```

This creates a dedicated experience for transportation providers.

21. Freight Quote Workflow

The complete quotation lifecycle can be represented as:

```
                FREIGHT JOB
                     |
             Notify Companies
                     |
        +------------+------------+
        |            |            |
    Company A    Company B    Company C
        |            |            |
      Quote        Quote        Quote
        +------------+------------+
                     |
                Review Quotes
                     |
               Accept Quote
                     |
            Assign Company
                     |
       Reject Remaining Quotes
                     |
             Start Shipment
```

This is one of the core business workflows of the platform.

22. Freight Data Architecture

The backend was structured around core freight entities. Conceptually:

```
Freight Job
     |
```

```
+-- Machine
+-- Pickup
+-- Delivery
+-- Required Date
|
+-- Quotes
       |
       +-- Freight Company
       +-- Amount
       +-- Status
```

This relational structure allows the system to maintain a complete history of the freight process.

23. REST API Architecture

The freight management platform uses REST APIs to connect the frontend with backend business logic. API functionality covers areas such as:

- Freight jobs
- Freight companies
- Company approval
- Quotes
- Quote acceptance
- Quote rejection
- Assignment
- Notifications
- Shipment status

The API architecture allows the frontend and backend to remain cleanly separated.

24. Backend Architecture

The backend was built using:

Node.js + Express.js + MongoDB + Mongoose

The backend handles:

- Business logic
- Authentication/authorization
- Freight workflows
- Quote processing
- Company management
- Notification triggers
- Database operations
- API responses
- Error handling

25. Automated Quote Processing

The quote workflow contains business automation rather than requiring every status change to be handled manually. For example:

```
Quote Accepted
      |
Update Selected Quote
      |
Assign Freight Company
      |
Reject Other Pending Quotes
      |
Trigger Notifications
      |
Update Freight Status
```

A single business action can therefore trigger multiple related system operations.

26. Production Deployment

The freight system was deployed as part of the production SYM Forklifts ecosystem. Infrastructure included:

- DigitalOcean
- Ubuntu
- Nginx
- PM2
- MongoDB
- GitHub Actions
- CI/CD

The production environment was configured to support the backend APIs and frontend application.

27. CI/CD

GitHub Actions was used as part of the deployment workflow.

```
Code Changes
     |
    Git
     |
  GitHub
     |
GitHub Actions
     |
Deployment
     |
Production Server
     |
Nginx + PM2
```

This provided a consistent deployment process for production changes.

Performance & Automation

200+ Freight Quotes Daily

The platform was designed to automate and manage **200+ freight quotes daily**, reducing manual administrative work involved in collecting and processing quotations.

Real-Time Shipment Visibility

The system provides shipment tracking/status visibility across the freight lifecycle.

Automated Communication

Email and SMS workflows reduce manual follow-up.

Automated Quote Processing

Accepting one quote can automatically update the remaining quotations.

Key Technical Challenges

Challenge 1 — Multiple Companies Quoting One Job

A single freight job can receive multiple quotations.

Solution: Designed quote records around the relationship between freight jobs and freight companies, allowing multiple companies to participate independently.

Challenge 2 — Quote Acceptance Logic

Once a quote is accepted, other quotes should no longer remain active.

Solution: Implemented automated quote-state updates so unsuccessful pending quotes can be rejected when a winning quote is accepted.

Challenge 3 — Automated Communication

Freight operations require timely communication.

Solution: Integrated Nodemailer and ClickSend to automate email and SMS notifications around important workflow events.

Challenge 4 — Machinery-Specific Freight

Industrial machines require more information than standard shipments.

Solution: Integrated machine specifications and transportation details directly into freight jobs.

Challenge 5 — Complex Freight Lifecycle

A shipment passes through several stages.

Solution: Implemented structured freight statuses and workflows covering job creation, quotation, assignment, pickup, transit, and delivery.

Complete Freight Workflow

The complete platform workflow can be summarized as:

```
                    FREIGHT REQUEST
                           |
                    CREATE JOB
                           |
                ADD MACHINE DETAILS
                           |
             ADD PICKUP & DELIVERY
                           |
                 PUBLISH FREIGHT
                           |
              NOTIFY FREIGHT COMPANIES
                           |
                 RECEIVE QUOTATIONS
                           |
                  REVIEW QUOTES
                           |
                  ACCEPT QUOTE
                           |
             +-------------+-------------+
             |                           |
     Assign Company            Reject Other Quotes
             |
       Notify Stakeholders
             |
           PICKUP
             |
         IN TRANSIT
             |
          DELIVERY
             |
        COMPLETED
```

My Contribution

I contributed across the full development lifecycle, including:

- Freight job management
- Machine specification integration
- Pickup and delivery workflows
- Freight company registration
- Company approval system
- Freight job browsing
- Quote submission
- Quote management
- Quote acceptance
- Automatic quote rejection
- Freight company assignment
- Email notification workflows
- SMS notification workflows
- Automated reminders
- Shipment status management

- Tracking functionality
- Google Maps integration
- Node.js / Express APIs
- MongoDB data modeling
- REST API development
- Production deployment
- Nginx configuration
- PM2 process management
- DigitalOcean infrastructure
- GitHub Actions / CI/CD

Final Outcome

The Freight Management System transformed a manual machinery transportation workflow into a centralized digital logistics platform. The system connects:

Freight Jobs + Machine Data + Freight Companies + Quotes + Assignment + Notifications + Tracking + Delivery

Instead of coordinating freight through disconnected emails, phone calls, and spreadsheets, the platform provides a structured workflow where freight companies can register, discover available jobs, submit quotes, and manage their transportation assignments.

At the same time, administrators can manage freight jobs, compare quotations, accept the appropriate quote, automatically reject unsuccessful quotes, monitor shipment progress, and keep stakeholders informed through automated email and SMS notifications.

Project Impact

200+	Freight quotes automated daily
Real-Time	Shipment status/tracking visibility
Automated	Quote acceptance and rejection workflows
Email + SMS	Automated stakeholder notifications
End-to-End	Freight job lifecycle management